

Kill Sheet — Subsea

The IWCF Subsea BOP kill sheet, with choke-line friction and initial dynamic casing pressure

⚠ FOR UNDERSTANDING PURPOSES ONLY

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Flutter / Dart

IWCF

Subsea BOP

9 live calculations

2-page PDF

1. What It Is

The Subsea Kill Sheet is the deepwater counterpart of the Surface Kill Sheet. It follows the IWCF “Subsea BOP Vertical Well Kill Sheet” in API field units and does everything the surface version does, then adds the subsea-specific items below.

Standard	IWCF Subsea BOP, vertical well, API field units
Reached from	Operations › Drilling › Well Control › Kill Sheets › Subsea
Saving	Five named slots in the app database
Output	Two-page A4 IWCF-formatted PDF through the Share menu

Two deepwater complications

- The **choke line** runs from the seabed BOP to the surface. Its friction (**CLF**, choke-line friction) raises the back-pressure the driller sees at the choke manifold.
- The **marine riser** above the BOP holds its own fluid column, so its volume has to be tracked as a separate row.
- The **initial dynamic casing pressure (IDCP)** corrects for choke-line friction so the driller sets the right choke pressure when starting to circulate.

The extra concept: IDCP

On a subsea well, the casing pressure gauge doesn't show the true formation-related pressure, because choke-line friction distorts it. $IDCP = SICP - CLF$ removes that distortion and is the pressure to hold at the choke when first opening up to circulate. Using SICP instead would over-pressure the formation by the CLF amount.

2. The Eight Sections

One more than the Surface sheet. Items marked ★ are subsea additions.

#	Section	Contents
1	Formation Strength Data	Formation strength test (A), mud weight at test (B), maximum allowable mud weight (C) and initial MAASP
2	Current Well Data	★ Marine riser length, ★ choke-line length, fluid density, casing shoe and hole data, and a subsea well diagram
3	Pump Data	Displacement, speed, ★ riser pressure and ★ choke-line pressure columns, ★ CLF (automatic)
4	Volumetric Data & Pump Strokes	Drill string, open-hole annulus, casing annulus, ★ choke-line volume row, surface volume and ★ marine riser volume row, with totals
5	Kick Data & Kill Mud Weight	SIDPP, SICP, pit gain and KMW (automatic)
6	Circulating Pressures	DPL, ICP, FCP and K100
7 ★	Initial Dynamic Casing Pressure	IDCP = SICP – CLF (automatic), a subsea-only section
8	Drill Pipe Pressure Step-Down	A schedule and line chart

3. How to Use It

1. Enter the **formation strength** data.
2. Enter the **marine riser** and **choke-line** lengths, fluid density and the casing and hole data. The subsea well diagram updates.
3. Enter pump displacement and speed, then **both the riser pressure and the choke-line pressure** at the slow rate. **CLF** appears.
4. Fill in the volume rows, including the choke-line capacity and the marine-riser capacity.
5. Enter **SIDPP, SICP and pit gain**. KMW appears.
6. ICP, FCP and K100 update. **IDCP** appears in its own section.
7. The step-down schedule and chart generate themselves.
8. **Save, Open or Export PDF**.

4. The Nine Calculations

Calculation	How it is worked out
1. Volumes	Length × capacity for each row, as on Surface, plus ★ choke-line volume (choke-line length × choke capacity) and ★ marine riser volume (riser length × riser capacity)

Calculation	How it is worked out
2. Strokes and times	Strokes = volume ÷ displacement, time = strokes ÷ speed, both rounded up . Extra pairs for the choke line and the marine riser
3. Max allowable mud weight and MAASP	Identical to Surface; maximum mud weight rounded down to 0.1 ppg, MAASP rounded down
4. Kill mud weight	Identical to Surface; rounded up to the next 0.1 ppg
5. Choke-line friction (CLF) ★	Choke-line pressure – riser pressure, both measured at the slow rate
6. DPL	The riser pressure at the slow rate, not the choke pressure
7. ICP	DPL + SIDPP
8. FCP and K100	FCP = (KMW ÷ current mud weight) × DPL; K100 is the pressure drop per 100 drill-pipe strokes
9. IDCP ★	SICP – CLF

The step-down schedule is identical to Surface: from ICP, drop by K100 every 100 strokes, never below FCP.

5. Saving

Like the Surface sheet, it uses the app database with **five named slots**, via its own dedicated Subsea set of save, open, list and delete actions. The slot label defaults to the sheet date. Opening a slot refills every field and re-runs all nine calculations.

6. The PDF

A two-page A4 IWCF-formatted document. It holds more than sixty values, including the subsea additions: up to four slow-rate pressure sets per pump, each with riser pressure, choke pressure and friction, the riser and choke-line lengths, the choke-line and marine-riser volume blocks, and the IDCP and CLF results.

7. Surface vs Subsea at a Glance

Feature	Surface	Subsea
Sections	7	8 (adds Initial Dynamic Casing Pressure)
Live calculations	8	9 (adds CLF and IDCP)
Pump pressure columns	One	Riser and choke line
Volume rows	Standard	Adds choke line and marine riser
Well data	Casing and hole	Adds riser and choke-line lengths

Feature	Surface	Subsea
Saving	Five slots	Five slots

Important note

Subsea kill sheet operations in a real well-control event must be performed by trained, IWCF- or IADC-certified personnel following the operator's procedures and the applicable deepwater regulations.